

are six layers of which top and bottom layers are usually made of protective glass. When electrons and holes are injected in the emissive layer, the light gets produced due to the process of electroluminescence.

For better performance, the chosen electrode must have a lower work function compared to the active organic layer when used for injection of electrons and vice versa for hole injection. For high work function, materials like indium tin oxide (ITO) are used while aluminium, calcium, or magnesium are the most common low work function options. Also, a device with large thickness leads to resistance against carrier injection, thereby decreasing carrier mobility and power efficiency. Although the internal quantum efficiency of OLEDs is high, the material of the electrode meant for electron injection must be transparent to visible light for external

quantum efficiency.

The main application of OLEDs is as a display in devices like televisions, tablets, and laptops. Panasonic, LG, Sony, and Philips are some of the prominent companies in this field. Active-matrix OLEDs (AMOLEDs), popular in smartphones and tablets, add an extra film of semiconducting material and are more flexible than OLEDs.

Recently launched smartwatch by Nubia sports a large flexible OLED screen that curves halfway round a person's wrist. Virtual reality (VR) headsets are also now using OLED micro-displays as these provide decreased brightness levels and high refresh rates.

Organic semiconductors are also used in the sustainable production of solar cells. Organic solar cells ensure easy fabrication and offer a low-cost alternative to traditional crystalline silicon-based cells. These are based

on the principle of photovoltaic effect. Organic polymers or small molecules present in these cells absorb sunlight and transport charge to produce electricity. These can store large amounts of solar energy due to large surface area and high absorption coefficient.

Research is ongoing to combine new materials while developing organic semiconductors for achieving better efficiency in applications. In August, a team of engineers from the Queensland University of Technology created a semiconductor capable of producing transistors, solar cells, and other electronic circuits by combining organic hydrocarbon naphthalene with a pigment called DPP, which is commonly used in Ferrari paint. Further advancements in the design process and adoption by emerging economies offer a promising future for organic semiconductors.

—Ayushee Sharma

3 THE IMPORTANCE OF SMART SPEAKERS IN MODERN ERA

Smart speakers are gaining huge popularity due to several new features that have made our lives easier and trouble-free. With just a command, smart speakers can now answer questions, set timers, play music, control other devices in the home, and also help children with their homework

With the advent of technology and electronic gadgets, the world has become smarter. Technology has become a part and parcel of our daily lives, solving the intricacies of daily routines and making things trouble-free. Smartphones are considered to be one of the biggest innovations in the field of technology.

These have made our lives easier. Another popular innovation after smartphones is smart speakers. These have gained huge traction around the globe ever since the first Amazon Echo speaker was launched in 2014. With just a command, smart speakers can now answer questions, set timers, play music, control other devices in the home, and also help children with their homework.

According to a survey by Canalys, over 17.1 million smart speakers were shipped globally in 2018. To offer a tough competition to Amazon, their biggest rivals Google and Apple came

up with their own voice assistant smart speakers such as Google Mini and Apple Homepod. Slowly, other companies like Marshall and Bose also entered the market.

Why do we need smart speakers? Smart speakers are slowly becoming an important part of every household. These can solve several problems on just a single command. For example, if you say, "Google, I am home," your smart speaker can turn on lights or TV.

Smart speakers and connected devices open the gate of possibilities for people who are dealing with limited accessibility, low vision, and more. Now, instead of physically switching on, you can command your voice assistant Alexa to run a hot shower for you. Tasks have become easier and trouble-free due to the introduction of smart speakers.

Which is the best voice assistant speaker? Apart from Alexa and Siri, there are many other voice assistants

available in the market such as Microsoft's Cortana, Panasonic's Arbo, and Samsung's Bixby. Earlier, Microsoft had partnered with Samsung to launch a speaker with its own voice assistant Cortana, but it, unfortunately, failed to gain traction in the market.

Among all, Amazon and Google have huge trucks of data that can improve the voice processing systems and interaction. Most importantly, Google Assistant is furnished with human-like natural language processing. However, Apple's Siri sometimes fails to understand people, mostly Indian accents.

Google Home. It is comprehensive and human-like in its capabilities and responses. Works well with existing Google products such as Pixel phones and Chromecast line.

A highlight of Google-based smart speakers is that these allow you to send content to Google Chromecast dongles as well as control playback using your voice. It's also very good at